

Thesis Proposal

Machine Learning Techniques for Baseline Covariate Adjustment
in Structural Equation Modeling

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1 Introduction

Various cross-lagged panel models (CLPMs) have been widely used in psychology to study the relationship between two variables over time (Usami et al., 2019). Suppose a researcher is interested in the relationship between a football player’s performance (Y_t , see Figure 1) and their self-confidence (X_t). The researcher suspects that good performances boosts a player’s confidence, which in turn enhances future performance. The researcher aims to interpret these relationships causally, requiring control for both time-varying and baseline confounders (C), for example, age or genetic makeup. Such baseline confounders lead to biased estimates when not, or inappropriately, controlled for.

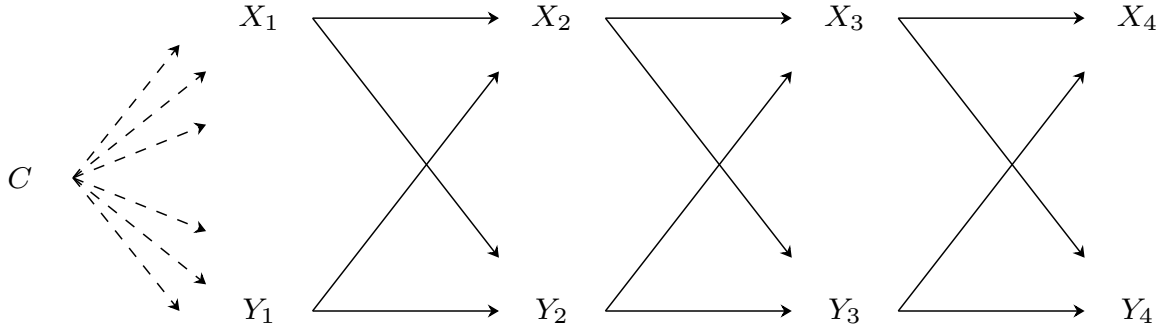


Figure 1: A DAG illustrating the relationship between self-confidence (X_t) and performance (Y_t), where C is the set of baseline confounders.

Several methods adjust for baseline confounding, such as the random-intercepts CLPM (Hamaker et al., 2015) and various dynamic panel models (Dishon and DeShon, 2022). A recent approach, baseline covariate adjustment (BCA) structural equation modeling (SEM) (Irmer et al., 2021), decouples control of time-invariant confounders from the estimation of the cross-lagged effects. BCA scores — residuals from regressing variables of interest on baseline covariates — replace observed variables in the SEM. A major strength of this approach is that the relationship between the variables of interest and the confounders can be flexibly modeled using machine learning (ML) techniques, such that the form of the adjustment function does not need to be assumed but can be learned from the data. In case of our example researcher, age might non-linearly affect the player’s self-confidence. BCA SEM can reduce model misspecification and in principle, lead to more robust causal inference.

However, implementations of ML techniques with BCA SEM do not exist yet. In this thesis, the impact of violations of the linearity assumption on causal estimates from CLPMs are investigated, and various ML techniques are implemented and compared, yielding the following research questions:

1. *How do violations of the assumption of linearity affect the performance of various CLPM methods?*
2. *How can ML techniques best be integrated with the BCA SEM approach to control for non-linear confounding effects?*

Unfortunately, a naive application of ML techniques does not guarantee a bias reduction. Their performance depends on tuning choices that trade off bias and variance, where variance can matter significantly in practice when resources are limited. Moreover, many ML methods do not directly provide standard errors, limiting their usefulness for inference (Keele and Grieve, 2025). In addition, using estimated adjustment functions inside the SEM can introduce plug-in bias; heavy smoothing can remove real signal (oversmoothing), and very flexible learners can chase noise (overfitting). Taken together, it is difficult to state in advance to what extent ML will make causal estimates from BCA SEM more robust. However, previous work using ML approaches has yielded flexible yet interpretable results in a social science context (e.g. Hainmueller and Hazlett, 2014), highlighting their potential to reduce misspecification bias.

2 Analytical Strategy

To answer our research question, we make use of simulations, specifying a data-generating model describing how two key variables linearly relate to each other over time. Effect sizes for the cross-lagged and autoregressive paths are chosen to reflect values commonly seen in the social sciences. Multiple confounding variables with a nonlinear effect on the main variables are included.

2.1 Conditions

Nonlinear patterns are introduced using (e.g.) polynomial terms or splines, with the exact functional forms to be decided later. We vary the amount of nonlinearity, for example, by changing the polynomial parameters. The number of confounders, measurement occasions, and sample sizes are also varied to reflect typical studies in social science contexts. BCA scores are first obtained under a linear model. We move forward with a minimal implementation of one ML method and expand to multiple techniques if time allows. Similarly, we will assess the effects of sample size, confounders, and measurement occasions after the minimal implementation is developed (see Table 1).

Table 1: Conditions and outcomes.

Conditions	Outcomes
<i>Confounder functional form</i>	Bias of cross-lagged parameters
<i>Extent of non-linearity</i>	Empirical s.e.
Number of confounding variables	Coverage
Sample size	Convergence
Number of measurement occasions	

2.2 Outcomes

Bias, empirical standard errors, coverage and convergence are estimated for the cross-lagged parameters using the BCA SEM framework.

2.3 Software

Simulations are run in R 4.5.1 (R Core Team, 2021). We use packages for different ML techniques (tbd.), such as ranger 0.17.0 (Wright and Ziegler, 2017) or xgboost 3.1.0.1 (Chen et al., 2025). Cross-lagged panel models are fitted with lavaan 0.6-20 (Rosseel et al., 2025).

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